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*Scenario: The analytics team of a food-ordering app website tracked 3,150 sessions last month through four stages: traffic source, landing page, next action and outcome. The Sankey diagram below shows the number of sessions on every flow. Three flows are hidden and marked "?".*

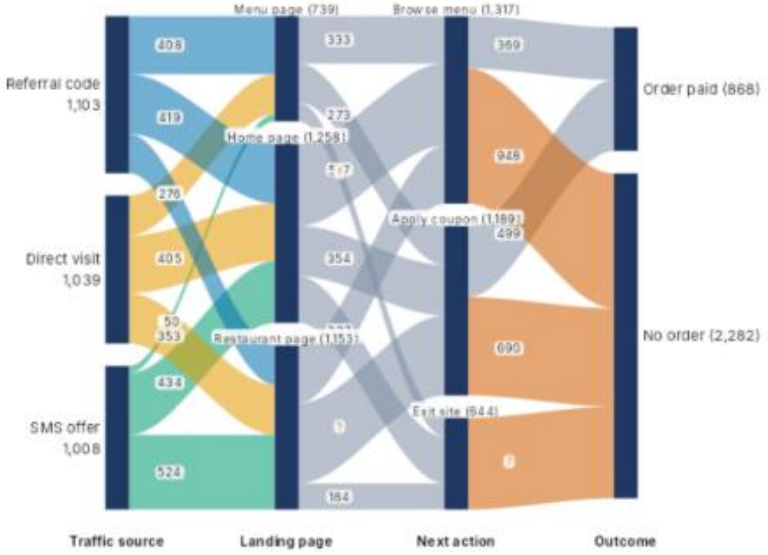



**Figure 1: Sankey diagram of website sessions (traffic source → landing page → next action → outcome)**

**(a) Using conservation of flow (what enters a node equals what leaves it) find the three hidden flows, showing working: Menu page to Exit site; Exit site to No order; Restaurant page to Apply coupon. State the rule you used.**

**Rule used:** Conservation of flow – for every intermediate node, total sessions entering = total sessions leaving = the node value shown in the diagram.

**1. Menu page → Exit site.** Menu page (739) leaves through Browse menu (333), Apply coupon (273) and Exit site:

Menu page → Exit site = 739 − 333 − 273 = 133 sessions

**2. Exit site → No order.** Order paid (868) is fully explained by Browse menu → Order paid (369) + Apply coupon → Order paid (499) = 868, so nothing reaches Order paid from Exit site. All Exit site sessions therefore go to No order:

Exit site → No order = 644 sessions (check: No order = 948 + 690 + 644 = 2,282 ✓)

**3. Restaurant page → Apply coupon.** Apply coupon (1,189) receives Menu page 273, Home page 354 and Restaurant page:

Restaurant page → Apply coupon = 1,189 − 273 − 354 = 562 sessions

Check from the Restaurant page node: 1,153 − 407 (to Browse menu) − 184 (to Exit site) = 562 ✓. Check on Exit site: 133 + 327 + 184 = 644 ✓.

**Answer: Menu page → Exit site = 133; Exit site → No order = 644; Restaurant page → Apply coupon = 562.**

**Table 1: All flows in the diagram (sessions)**

| From | To | Sessions | Basis |
|---|---|---|---|
| Referral code | Menu page | 408 | Label in diagram |
| Referral code | Home page | 419 | Label in diagram |
| Referral code | Restaurant page | 276 | Label in diagram |
| Direct visit | Menu page | 281 | Balance of node totals |

| From | To | Sessions | Basis |
|---|---|---|---|
| Direct visit | Home page | 405 | Label in diagram |
| Direct visit | Restaurant page | 353 | Label in diagram |
| SMS offer | Menu page | 50 | Label in diagram |
| SMS offer | Home page | 434 | Label in diagram |
| SMS offer | Restaurant page | 524 | Label in diagram |
| Menu page | Browse menu | 333 | Label in diagram |
| Menu page | Apply coupon | 273 | Label in diagram |
| Menu page | Exit site | 133 | Hidden – calculated |
| Home page | Browse menu | 577 | Balance of node totals |
| Home page | Apply coupon | 354 | Label in diagram |
| Home page | Exit site | 327 | Balance of node totals |
| Restaurant page | Browse menu | 407 | Balance of node totals |
| Restaurant page | Apply coupon | 562 | Hidden – calculated |
| Restaurant page | Exit site | 184 | Label in diagram |
| Browse menu | Order paid | 369 | Label in diagram |
| Browse menu | No order | 948 | Label in diagram |
| Apply coupon | Order paid | 499 | Label in diagram |
| Apply coupon | No order | 690 | Label in diagram |
| Exit site | No order | 644 | Hidden – calculated |

Every node balances: inflow = outflow for all landing pages and next actions; sources total 1,103 + 1,039 + 1,008 = 3,150 and outcomes total 868 + 2,282 = 3,150.

**(b) Compute: (i) the overall conversion rate, i.e. Order paid as a percentage of all sessions; (ii) the largest single flow into "No order" or "Exit site"; (iii) the percentage of "Restaurant page" sessions that go on to "Apply coupon".**

**(i) Overall conversion rate**

= Order paid ÷ total sessions × 100 = 868 ÷ 3,150 × 100 = 27.56%

**(ii) Largest single flow into "No order" or "Exit site"**

Into No order: Browse menu 948, Apply coupon 690, Exit site 644. Into Exit site: Menu page 133, Home page 327, Restaurant page 184.

**Largest = Browse menu → No order = 948 sessions** (948 ÷ 3,150 = 30.1% of all sessions).

**(iii) Restaurant page sessions going to Apply coupon**

= 562 ÷ 1,153 × 100 = 48.74%

**(c) Design the final Sankey: give four design decisions (node order to reduce crossings, colour rule, labels, what to highlight) and justify each in one line using this data.**

| Decision | Choice | Justification (using this data) |
|---|---|---|
| 1. Node order | Keep Order paid above No order; order next actions Browse menu → Apply coupon → Exit site and landing pages Menu → Home → Restaurant, placing each node near the weighted average position of its largest links (barycentre ordering). | The big flows Home → Browse (577), Restaurant → Apply (562) and Browse → No order (948) then run almost straight, and the small Exit-site flows (133, 184, 327) sit at the bottom, so crossings are minimal. |

| Decision | Choice | Justification (using this data) |
|---|---|---|
| 2. Colour rule | Colour each link by the node it leaves: three distinct hues for Referral code, Direct visit and SMS offer; neutral grey for later flows; green for the Order paid node and red for the No order node. | With only three sources, three hues stay distinguishable, and the green/red end nodes make gain versus loss obvious (868 paid vs 2,282 lost). |
| 3. Labels | Label every node with name, sessions and % of 3,150 (e.g. Order paid 868 · 27.6%); label only links of 300+ sessions and add the four stage headings. | 17 of the 23 flows are 300+ sessions; leaving the six small flows (50, 133, 184, 273, 276, 281) unlabelled avoids clutter while totals still add up. |
| 4. Highlight | Draw Browse menu → No order (948) in strong red with every other flow muted, and annotate the Apply coupon conversion. | It is the single biggest loss (30.1% of all sessions), and coupon users convert at 499 ÷ 1,189 = 42.0% versus 369 ÷ 1,317 = 28.0% for Browse menu – the key insight. |

**(d) The Sankey is to be printed with 1 cm of node height = 200 sessions. Compute the height in cm of each landing-page node (Restaurant page, Home page, Menu page).**

Height (cm) = sessions ÷ 200

| Landing-page node | Sessions | Height (cm) |
|---|---|---|
| Restaurant page | 1,153 | 1,153 ÷ 200 = 5.765 ≈ 5.77 cm |
| Home page | 1,258 | 1,258 ÷ 200 = 6.29 cm |
| Menu page | 739 | 739 ÷ 200 = 3.695 ≈ 3.70 cm |

Check: 5.765 + 6.29 + 3.695 = 15.75 cm = 3,150 ÷ 200 (node heights only, excluding the gaps between nodes).

**(e) Write R code: a links data frame for the first four flows leaving "Referral code" and "SMS offer" in the table order of the diagram, a nodes data frame, and the networkD3::sankeyNetwork() call (or plotly) that draws the whole diagram.**

```
# install.packages("networkD3")
library(networkD3)

# ---- Nodes: ids 0-10 in diagram order (top to bottom per stage) ----
nodes <- data.frame(
  name = c("Referral code", "Direct visit", "SMS offer",    # 0-2 source
           "Menu page", "Home page", "Restaurant page",    # 3-5 landing
           "Browse menu", "Apply coupon", "Exit site",     # 6-8 next action
           "Order paid", "No order"),                       # 9-10 outcome
  stringsAsFactors = FALSE
)

# ---- Links: first four flows leaving Referral code and SMS offer ----
# (3 flows from Referral code, then the first from SMS offer)
links_first4 <- data.frame(
  source = c(0, 0, 0, 2),     # Referral x3, SMS
  target = c(3, 4, 5, 3),     # Menu, Home, Restaurant, Menu
  value  = c(408, 419, 276, 50)
)

# ---- Remaining flows, so the call draws the WHOLE diagram ----
links_rest <- data.frame(
  source = c(1, 1, 1,  2, 2,  3, 3, 3,  4, 4, 4,  5, 5, 5,
             6, 6,  7, 7,  8),
  target = c(3, 4, 5,  4, 5,  6, 7, 8,  6, 7, 8,  6, 7, 8,
             9, 10,  9, 10,  10),
  value  = c(281, 405, 353,  434, 524,
             333, 273, 133,
             577, 354, 327,
             407, 562, 184,
```

```
              369, 948,
              499, 690,
              644)
)

links <- rbind(links_first4, links_rest)
# colour each link by its source node
links$group <- nodes$name[links$source + 1]

# ---- Colour rule: 3 source hues, grey middle, green/red outcomes ----
colours <- 'd3.scaleOrdinal()
  .domain(["Referral code","Direct visit","SMS offer","Menu page","Home page",
           "Restaurant page","Browse menu","Apply coupon","Exit site",
           "Order paid","No order"])
  .range(["#4E79A7","#F28E2B","#59A14F","#BDBDBD","#BDBDBD",
          "#BDBDBD","#BDBDBD","#BDBDBD","#BDBDBD",
          "#2CA02C","#D62728"])'

# ---- Draw the whole Sankey ----
sankeyNetwork(Links = links, Nodes = nodes,
              Source = "source", Target = "target", Value = "value",
              NodeID = "name", NodeGroup = "name", LinkGroup = "group",
              colourScale = JS(colours),
              units = "sessions",
              fontSize = 13, nodeWidth = 25, nodePadding = 12,
              iterations = 0,   # keep the node order given above
              sinksRight = FALSE,
              width = 900, height = 500)
```

**(f) State two situations in which a Sankey diagram misleads or becomes unreadable, and name one alternative chart for each.**

| No. | Situation where a Sankey misleads / becomes unreadable | Alternative chart |
|---|---|---|
| 1 | Too many nodes and links (e.g. dozens of pages) – flows cross heavily and thin links become untraceable spaghetti. | Heatmap of a source × destination matrix |
| 2 | The task is to compare exact flow sizes or drop-off between steps – judging ribbon widths and angles is imprecise, and tiny flows such as 50 vs 133 are almost invisible. | Sorted (grouped) bar chart or funnel chart |